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ME 498 Opticon Report

I. Abstract

Topology optimization is a common tool to optimize the geometry of engineered components in question. This project utilizes a SIMP topology optimization formulation to reduce the compliance of a bike frame structure, while retaining sufficient mass. The final designs retained 30% of the maximum mass in the design domain, while decreasing all compliances to the negative second and third orders of magnitude. These optimizations demonstrate stiffer bike frame designs while retaining structural rigidity.

II. Introduction



Figure 1. Biker 'cruising' on the Main Quad at UIUC

Especially for university students, biking has become one of the most popular methods of transportation. One of the most engineered components of a bike assembly is the main frame, supporting the rider and connecting all the components together.

As engineering methods and demand for bicycles develop, the need for frames fitting riders' various specific needs becomes increasingly important. Achieving a design trading off various factors such as stiffness and mass remains a central challenge to bike frame design.



Figure 2. Traditional bike design and terminology

Traditional bike design involves 6 main components, as shown in Figure 2. This general design has proven effective in balancing manufacturability and general use cases, but with modern engineering tools, such as topology optimization, perhaps more sophisticated and optimized frame designs can be considered.

Thus, this report explores applying topology optimization to the frame of a bicycle. Through implementing this optimization, we can obtain frame design minimizing the compliance, providing a more robust biking experience, while maintaining necessary structure. While not a perfect representation of a finalized product, this report will provide insight into a more modern and specialized perspective on bike frame design.

III. Methods

a. Assumptions

In order to simplify the engineering problem to be more feasible and efficient for the numeric solver, we must apply certain key assumptions:

i. The frame is treated as 2D. Although the frame generally splits off to support the rear wheel, we will treat the thickness in this portion as twice the thickness of the frame portions that split off (i.e. seat and chain stays).

ii. All forces are direct loads onto the nodes nearest to the seat, crank, and handlebars. In reality, these forces would be moment loads transmitted from the seat, handlebars, and crankset, but for the purposes of this topology optimization design, we will treat this as direct loads. To expedite the optimization, we choose the dominant loading factors, and these direct loads are the primary effects from riding. For the same reason, we also will

not consider the force from the chain tension, as that is considered to be far less dominant than the direct loads from the rider.

iii. All forces are within the 2D plane. Generally, as the rider oscillates transmitting force on the right and left side, the net load produces a force 'out of the page' or 'into the page,' but we will consider the average case where these loads cancel out, so loads are only vertical or horizontal.

iv. The rear and front wheel will be treated as pin and roller constraints, respectively. In our frame of reference, this assumes the rear dropout is fixed, while the front dropout is free to move parallel to the riding surface.

v. All optimizations will be performed using aluminum. Bike frames vary in material, but aluminum is a common material for most bikes purchased by average consumers.

b. Load and Boundary Condition Configurations



Figure 3. Cruising (left), climbing (middle), braking (right)

Based on how a rider is using their bike, the loading configuration will change. Thus, we will consider 3 basic riding conditions: cruising (Configuration A), off-saddle climbing (Configuration B), and braking (Configuration C). To simulate these 3 riding conditions, we will vary the forces on the seat, handlebars, and cranks, given the assumptions listed above (Table 1). Note that although there is generally no load on the seat during climbing, 25 N is used to enforce material in that area. Figure 4 shows the FBD for the loading used in this model.



Figure 4. Free body diagram of the bike frame (coordinates for load/BC points shown)

Table 1. Force values for cruising, climbing, and braking load configurations			
Value	A. Cruising	B. Climbing	C. Braking
$F_{crank,x}$ [N]	150	200	500
$F_{crank,y}$ [N]	-500	-1500	-500
$F_{seat,x}$ [N]	0	0	0
$F_{seat,y}$ [N]	-500	-25	-200
$F_{bar,x}$ [N]	-100	-100	-1000
$F_{bar,y}$ [N]	-100	-100	-200

The relative loads shown in Table 1 were influenced by common practice relative magnitudes [1][2] in other academic structural studies if bicycle frames.

c. Optimization Statement

As noted in the introduction, this optimization will minimize the compliance of the frame, with the main constraint being the maximum mass allowed. Deflection and stress were initially tested as constraints, but testing has shown that using only the mass constraint sufficiently keeps deflection and stresses in reasonable ranges while also running significantly faster. Thus, the optimization problem can be written as:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^{\boldsymbol{n}}} \quad & C(\mathbf{x}) = \boldsymbol{f}^T\mathbf{u}(H\mathbf{x}) \\ s.t. \quad & K(H\mathbf{x})\mathbf{u} - \boldsymbol{f} = \mathbf{0}, \\ & \textstyle\sum_{\boldsymbol{e=1}}^{\boldsymbol{n_e}}(\rho t A_e \cdot Hx_e) - \mathbf{0.3}m_{\boldsymbol{max}} \leq \mathbf{0}, \\ & 10^{-3} \leq x_e \leq 1.0, \end{aligned}$$

where:

$\mathbf{x}$ is a vector of n element densities,

H is the hat-filter matrix,

$C(\mathbf{x})$ is the compliance value,

$\boldsymbol{f}$ is the force vector,

$\mathbf{u}(H\mathbf{x})$ is the displacement vector,

$K(H\mathbf{x})$ is the SIMP stiffness matrix,

ρ is the material density, t is the plate thickness, A_e is the element area,

and $m_{\boldsymbol{max}}$ is the maximum area if the entire design domain were filled.

With the SIMP material interpolation, $K(H\mathbf{x})$ is defined with

$$E_e = E_0(H\mathbf{x})^p,$$

where the penalty term $p = 3$.

Thus, we classify this problem as nonconvex (from the SIMP $p > 1$), nonlinear (also from the SIMP $p > 1$), and continuous ($x_e \epsilon([10^{-3}, 1] \cap \mathbb{R})$). Moreover, this is a (density method) topology optimization because we fix the outer design domain and vary the density field among the entire domain. It is also of note that if we considered the design variable to be the penalized $(H\mathbf{x})^p$.

d. Sensitivity Derivations

When conducting (SIMP) topology optimization, we must know the sensitivities of the objective and constraint, or the derivatives of the objective and constraint with respect to the design variable.

The following is the derivation for the compliance (objective) sensitivity:

We start with the state equation:

$$R = K(x)u - f(x) = 0.$$

The total derivative is:

$$\frac{df}{dx_i} = \frac{\partial f}{\partial x_i} - \varphi^T \frac{\partial R}{\partial x_i},$$

where:

$$\varphi = -\left(\frac{\partial R}{\partial u}\right)^{-T} \frac{\partial C}{\partial u}^T.$$

The explicit dependence of compliance on the design variable is:

$$\frac{dC}{dx_i} = \frac{\partial f}{\partial x_i}^T u + f \frac{\partial u}{\partial x_i}.$$

The explicit dependence of compliance on the state vector is:

$$\frac{\partial C}{\partial u} = f^T.$$

The explicit dependence of the state equation on the state vector is:

$$\frac{\partial R}{\partial u} = K.$$

The explicit dependence of R on the design variable is:

$$\frac{\partial R}{\partial x_i} = \frac{\partial K}{\partial x_i} u + K \frac{\partial u}{\partial x_i} - \frac{\partial f}{\partial x_i}.$$

Substituting everything into the total derivative, we obtain:

$$\frac{dC}{dx_i} = -u^T \frac{\partial K}{\partial x_i} u,$$

which is the compliance sensitivity.

The following is the derivation for the mass (constraint) sensitivity:

Total mass is:

$$\boldsymbol{m} = \sum_{\boldsymbol{i=1}}^{\boldsymbol{n_i}} (\rho t A_e \cdot x_i).$$

This is linearly dependent on element-wise density x_i, so we obtain:

$$\frac{dm}{dx_i} = \sum_{\boldsymbol{i=1}}^{\boldsymbol{n_i}} (\rho t A_e),$$

which is the mass sensitivity.

e. Python Setup

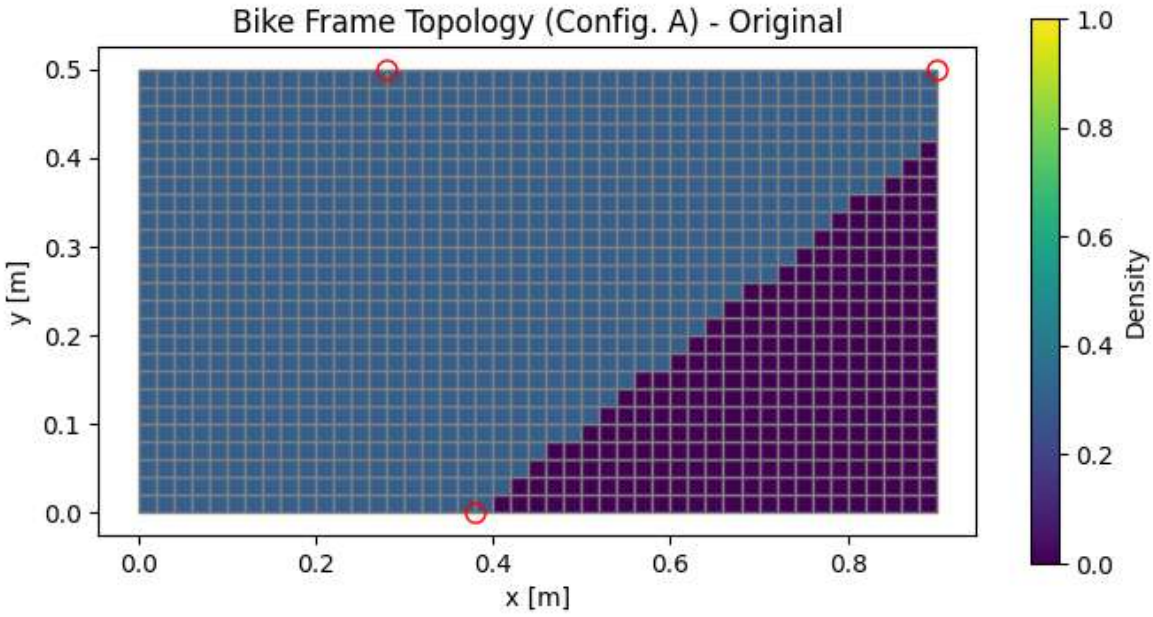


Figure 5. Design domain and initial configuration (Config. A); red circles: loads

Figure 5 shows the design domain and initial densities for the topology optimization (all configurations have the same initial condition, but only configuration A is shown). A passive void

region is used to avoid placing material where the front wheel is. The initial condition is a uniform grid with maximum mass ($x_i = 0.3$). Additionally, the mesh grid size is (45, 25) (including the passive void region). Moreover, as seen in the Appendix, we use the Scipy SLSQP minimize method. Furthermore, Table 2 shows the filter radii used. Density filtering is also performed at each iteration to produce more realistically manufacturable results.

IV. Results

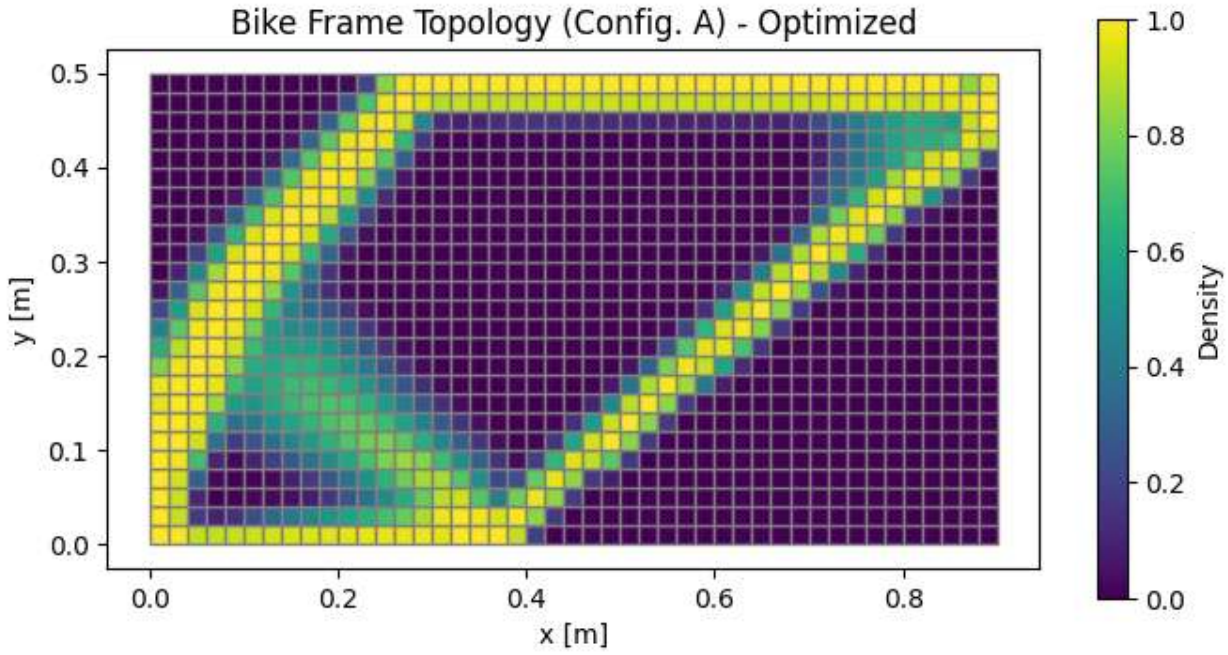


Figure 6. Configuration A topology optimization result (cost: $48.26)

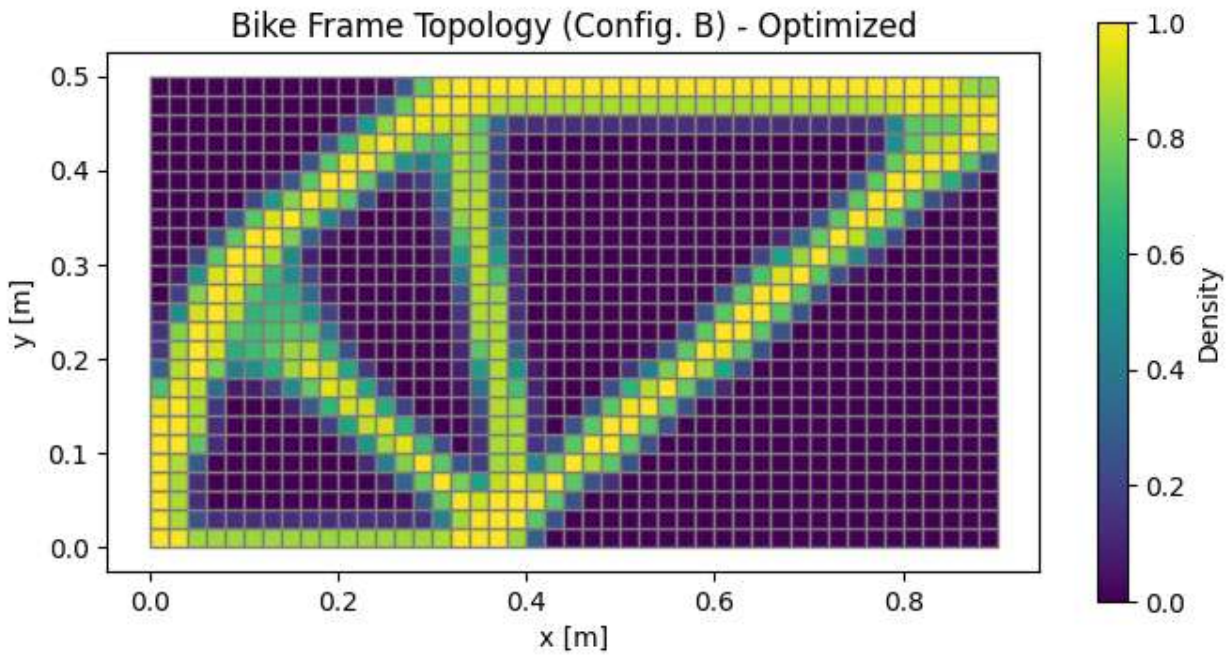


Figure 7. Configuration B topology optimization result (cost: $48.26)

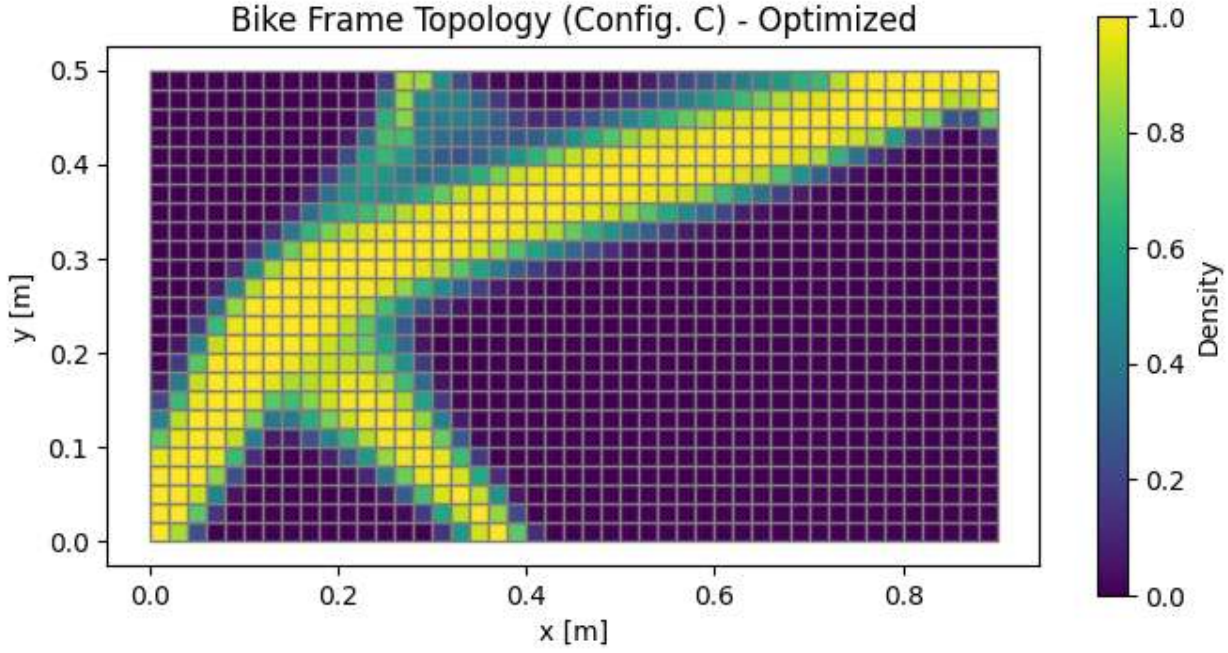


Figure 8. Configuration C topology optimization result (cost: $48.26)

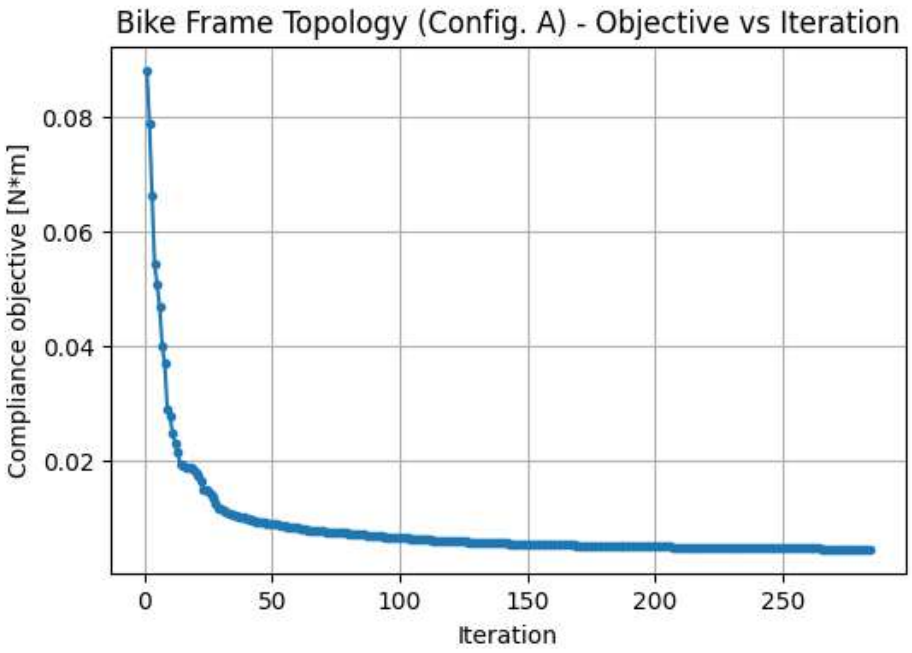


Figure 9. Configuration A topology optimization convergence plot

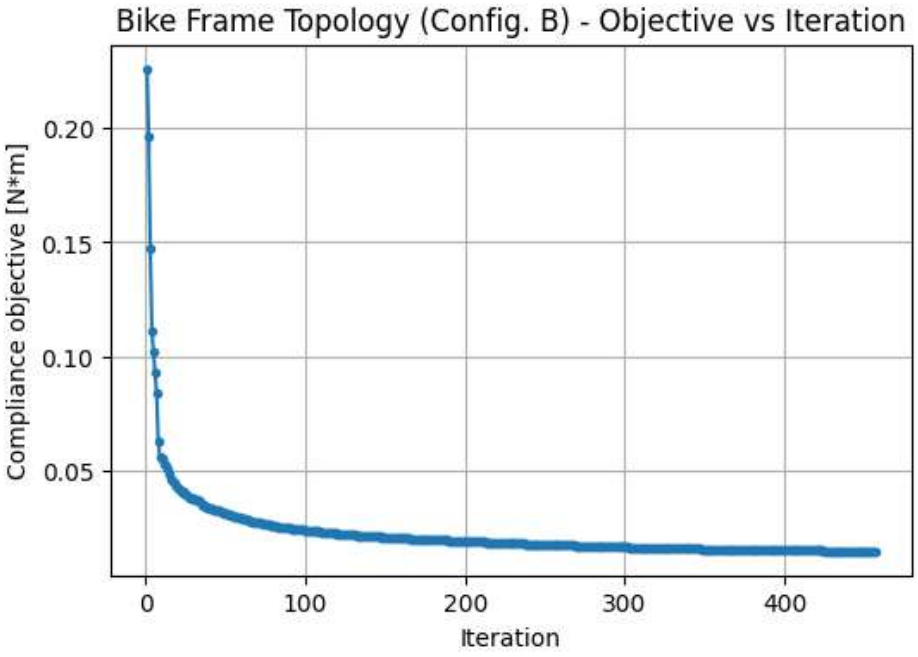


Figure 10. Configuration B topology optimization convergence plot

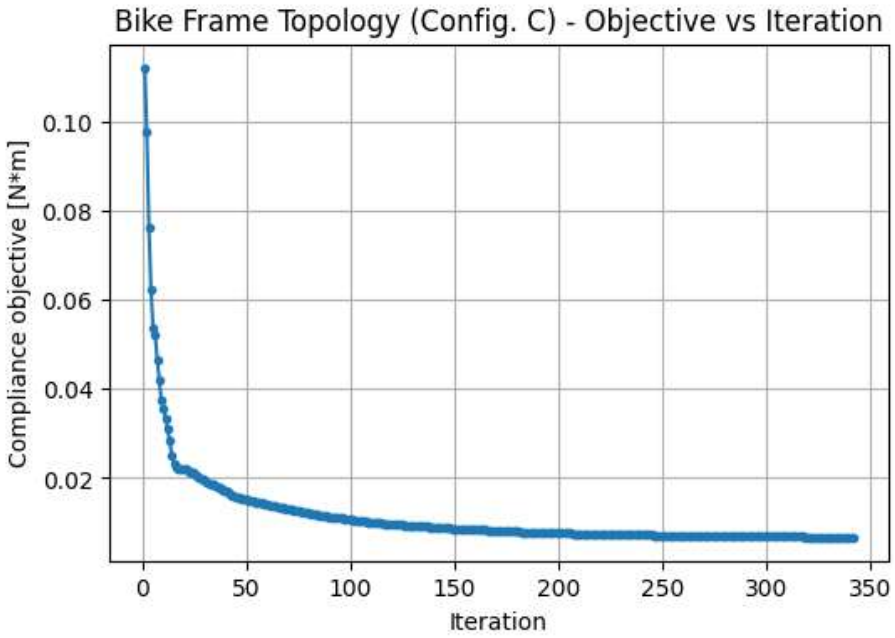


Figure 11. Configuration C topology optimization convergence plot

a. Configuration A

Of all the configurations, this seems to be the most similar to a typical 6-component bike frame design, except that the seat tube does not extend to the actual seat position. Additionally, there appears to be a dense outlining 'parallelogram' framing with a thinner beam running in the middle, connecting two of the sides. Moreover, it is interesting that the 'seat stay' area is significantly thicker than the other portions of the framing, suggesting higher loads at the rear end of the bike with this model. It is also of note that in a more realistic design, perhaps more space would be needed to make in the rear end so the chain can cross through the frame. However, this depends on gear diameters, which changes vertical chain position.

b. Configuration B

Configuration B is similar to Configuration A, but with a second cross-bracing member in the middle. This is most likely because of the significantly higher crank loading, as the vertical member stiffens the area around the crank. Additionally, all the members seem to be relatively uniform in width besides the bottom horizontal member connecting the crank and rear wheel. This suggests loads are relatively uniform in the top portions of the framing.

c. Configuration C

This output design appears to be the most novel compared to the other configurations simulated. In the real world, this design would not be made, but this does provide insight on bikes designed for conditions with high frequencies or magnitudes of braking. Particularly, the rear and top portions of the framing are nearly full density and much wider than the members seen in the

other frame designs, suggesting bikes designed more specifically for handling breaking should thicken those members.

d. Convergence and Comparison

Table 2. Final Compliances	
Configuration	$C\ [\times 10^{-3} N \cdot m]$
Cruising	4.62
Climbing	14.95
Braking	6.70

Table 2 shows the final obtained objective compliance values for the 3 optimization configurations. It is of note that the minimized compliance for climbing is an order of magnitude higher than the other two configurations, but all compliances are minimal.

Figures 9-11 show the convergence of the objective function (compliance) with respect to the optimizer iterations. From the convergence plots, we are confident that the optimizations found at least one optimal density field for each load configuration, as an asymptote is seen (and all optimization runs terminate successfully before the max iterations of 500). Additionally, all 3 designs utilized the maximum material cost ($48.2625, or 18.5625 kg Al, which is 30% of the maximum cost of the entire design domain). Table 3 shows the filter radii used to obtain the final density fields. The radii values were tuned using a coarser mesh before the final outputs were obtained.

Table 3. Density Filter Radii	
Configuration	r_{min} [m]
Cruising	0.023
Climbing	0.027
Braking	0.025

When looking at the results, we see that configurations A and B achieved much more similar final density fields than configuration C. Contextually, this is logical, as configurations A and B are when the rider is moving the bike, whereas configuration C is when the rider stops the bike. Moreover, configurations A and B seem significantly more similar to common bike designs, whereas configuration C lacks a downtube geometry.

V. Conclusions

To conclude, this project obtained optimal bike frames for 3 different load configurations. Although topology optimization is only one tool in the design process, we can use the results to

influence actual design and manufacturing of bike frames. Insights from these results can be powerful tools for influencing both novel and existing designs.

Continuing from this work, more configurations can be considered, and the current model can also be tweaked. For instance, more passive void regions (such as for the chain) may be considered, along with required material sections. Moreover, more complex loading can be considered, reducing the assumptions.

References

[1] D. Covill, P. Allard, J.-M. Drouet, and N. Emerson, “An assessment of bicycle frame behaviour under various load conditions using numerical simulations,” Procedia Engineering, vol. 147, pp. 665–670, 2016. [Online]. Available: http://184.163.55.35/iberg/2016_COVILL_An%20Assessment%20of%20Bicycle%20Frame%20Behaviour.pdf

[2] D. S. De Lorenzo and M. L. Hull, “Quantification of structural loading during off-road cycling,” *Journal of Biomechanical Engineering*, vol. 121, no. 4, pp. 399–405, 1999. [Online].
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